Manuel Zechmann

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Education

University of Vienna — Vienna, Austria

Master of Science (MSc), Media Computer Science, Grade Avg: 1.56 — *Mar 2023 – Nov 2025*

- Master's Thesis: Comparative Analysis of Procedural Planet Generators (GAME-ON 2025). Investigated algorithmic terrain synthesis, LOD optimization, and real-time rendering performance.
- Rigorous coursework spanning advanced computer graphics, image synthesis, computer vision, machine learning, and mathematical foundations of rendering.
- Acted as Team Lead in advanced projects, coordinating cross-functional collaboration and architectural design.

University of Vienna — Vienna, Austria

Bachelor of Science (BSc), Computer Science, Grade Avg: 2.32 — *Oct 2019 – Jan 2023*

- Bachelor's Thesis: Training a CNN to Detect Hands and Handposes. Designed a procedural data generation pipeline in Unity to train robust deep learning models.
- Solidified foundations in algorithms, data structures, Computer Vision, and Computer Graphics.

Publications

Comparative Analysis of Procedural Planet Generators

GAME-ON 2025, Ghent

- Published and presented research paper focusing on procedural planet generation and performance metrics.

Awards

UNIVIE Research Award for Students

University of Vienna, 2025

- Awarded for academic research and publication of the paper "Comparative Analysis of Procedural Planet Generators".

Leistungsstipendium

University of Vienna, 2025

- Received the Leistungsstipendium (Merit Scholarship) for top-tier academic performance.

Experience

Snap Inc. (Specs / Spectacles) — Vienna, Austria

Computer Vision Researcher — *Aug 2025 – May 2026*

- Built synthetic data generation systems using procedural generation, agentic workflows, MCPs, 3D Gaussian Splatting, and VLMs to create training data for vision models. Improved generated data with augmentations and methods to bridge the sim-to-real gap, like ControlNet.
- Trained and adapted vision models for 3D scene understanding, proving effectiveness of generated data with ablation studies.
- Engineered large-scale data generation pipelines on GCP (Vertex AI, GCS), generating up to 5 TB of structured data to train robust vision models.
- Collaborated closely with the team through reading groups and cross-functional discussions, and proactively supported team efforts.

ViewApp — Vienna, Austria

Multimedia Software Engineer — *Jun 2024 – Sep 2024, Jan 2025 – Jul 2025*

- Engineered custom C++ Unreal Engine tools and world-building pipelines, implemented snapping mechanics, and helped ship an Unreal Engine game to Steam, PlayStation 5, and Xbox.
- Led the technical implementation of the core interactive tutorial system, coordinating necessary cross-functional deliverables.
- Developed workflow automation tools and configured automated build processes using TeamCity to optimize production efficiency.

Partium — Vienna, Austria

Junior Software Engineer — *Oct 2022 – Mar 2024*

- Trained computer vision models for spare part recognition and integrated OCR pipelines for high-precision text extraction.
- Managed containerized microservices using Docker and Kubernetes in beta testing environments.

Partium Vienna, Austria

Software Engineer Intern *Jul 2022 – Sep 2022*

- Evaluated and integrated Elasticsearch as a high-performance search backend, accelerating large-scale dataset retrieval.

Projects

VulkanParticles Game Engine | C++, Vulkan, CMake, GoogleTest May 2026 – Present

- Architected and developed a high-performance C++ particle game engine utilizing the Vulkan API, designed for modular expansion into a full-scale engine.
- Implemented robust software engineering practices, featuring automated CI/CD pipelines and comprehensive unit testing with GoogleTest.

GPT from Scratch (nanoGPT) | Python, PyTorch, Transformers 2024

- Engineered a decoder-only transformer architecture from scratch in Python and PyTorch, based on Andrej Karpathy's reference implementation.
- Trained and evaluated the language model, gaining hands-on expertise in tokenization, self-attention mechanisms, and compute optimization.

Tall Trunk Galaxy 64 (ROM Hack) | C, Blender, SM64 SDK, Procedural Design Apr 2024

- Developed a small SM64 game utilizing the C source code and Blender asset workflows within strict N64 hardware constraints.

Skills

Languages: C++, Python, C, SQL, GDScript
Computer Vision & AI: PyTorch, OpenCV, Hugging Face, VLMs, ControlNet, Agentic Workflows (MCP), Synthetic Data Generation, Sim-to-Real / Domain Gap, Ablation Studies
Computer Graphics & 3D: OpenGL, Vulkan, GLSL/HLSL, Compute Shaders, Physically Based Rendering, Path Tracing, BVH, Unreal Engine, Godot, 3D Gaussian Splatting, Procedural Generation
Systems & Performance: CMake, Multi-threading, Profiling (Nsight, RenderDoc)
Cloud & DevOps: GCP (Vertex AI, GCS), Docker, Kubernetes, CI/CD, TeamCity, Git
Research Focus: Real-Time / Neural / Physically Based Rendering, Game Engines, 3D Scene Understanding, Synthetic Data, Generative Models, Multimodal / Vision–Language Models
Spoken Languages: German (Native), English (C2), Spanish (A2)